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Objective

Lead Software Engineer with 8+ years of experience designing, developing, and delivering scalable backend and full-stack applications, distributed systems, REST APIs, microservices, and event-driven architectures. Strong hands-on expertise in Node.js, TypeScript, JavaScript, Python, API integration, Kafka, RabbitMQ, Redis, Docker, Kubernetes, Terraform, CI/CD, and SQL/NoSQL databases. Proven experience in system design, performance optimization, cloud-native development, observability, technical leadership, mentoring, and cross-functional delivery. Experienced in building reliable, secure, high-performance systems and researching Agentic AI solutions for workflow automation and intelligent decision-making.

Experience

Lead Engineer : Air India , India

02/2025 - Present

- Architected and developed WhatsOnMyAI, an automated meal card parsing & data-extraction platform for Air India's F&B Team, eliminating manual operational work.
- Engineered custom LLM API workflows using a multi-model Agentic AI approach to streamline unstructured meal card PDFs into structured relational database schemas.
- Designed and implemented an internal Web Portal with Human-in-the-Loop (HITL) review capabilities, enabling F&B stakeholders to seamlessly verify and edit AI-processed menu data.
- Lead the design and development of critical APIs and distributed systems, including OCN Event Handling, Passenger Management APIs, and Payment Event Processing solutions.
- Architect scalable, event-driven backend systems for high-volume transactional workflows with strong emphasis on performance, reliability, fault tolerance, scalability, and availability.
- Collaborate with cross-functional teams to define system architecture, conduct technical reviews, translate requirements into technical solutions, and integrate new APIs with existing aviation systems.
- Mentor and guide junior and mid-level developers on API design, system architecture, clean code, software engineering best practices, and DevOps methodologies.

Senior Software Engineer : Phases India Technology Solutions Pvt Ltd , India

06/2019 - 12/24

- Led a team of 5 developers to design, develop, test, and deploy enterprise-grade applications using Node.js, TypeScript, MongoDB, REST APIs, and modern software engineering practices.
- Designed and implemented scalable backend services and distributed systems using microservices and event-driven architecture.
- Refactored and optimized legacy systems, achieving a 60% improvement in response time through Node.js worker threads and load-balancing strategies.
- Built distributed, event-driven and message-driven architectures using RabbitMQ and Kafka to support high-volume transactional workflows.
- Designed and implemented secure B2B API integrations and REST APIs connecting third-party platforms, internal services, and legacy systems.
- Implemented Infrastructure as Code using Terraform and automated cloud deployments, reducing infrastructure provisioning time by over 60%.
- Integrated CI/CD pipelines using GitLab, Docker, and Kubernetes to automate application delivery and improve deployment reliability.

- Designed and deployed centralized logging, monitoring, and observability using the ELK Stack, reducing average issue resolution time by 50%.
- Collaborated with technical and non-technical stakeholders to understand requirements, define technical solutions, align project goals, and deliver maintainable software.
- Participated in technical reviews, code reviews, engineering planning, troubleshooting, performance optimization, and continuous improvement initiatives.

Junior Software Engineer : Unieke Software Solutions , Kochi, India

01/2018 - 05/2019

- Worked as a full-stack developer on consumer-facing web applications using Laravel, Vue.js, Bootstrap, jQuery, JavaScript, and PHP.
- Developed a scalable web platform with real-time chat, profile filtering, role-based access control, and administrative dashboards.
- Implemented Progressive Web App (PWA) capabilities and OAuth 2.0 authentication to improve accessibility and application security.
- Built notification modules for user activity, messaging, and administrative alerts to improve application engagement and real-time interactivity.
- Automated code-quality enforcement using PHP CS Fixer and participated in peer code reviews to maintain engineering standards.

Projects

WhatsOnMyAI (Air India F&B Automation)

- Architected and built WhatsOnMyAI, an automated meal card parsing and verification system for Air India's F&B team. Streamlined unstructured meal card PDFs via a custom multi-model Agentic LLM API workflow into a relational database, and created an interactive portal featuring Human-in-the-Loop (HITL) review to eliminate manual data entry.
- **Tech Stack:** LLM APIs, Multi-Agent AI Framework, Node.js, Python, PostgreSQL, Human-in-the-Loop (HITL), React.js

Podio Backup Tool

- Designed and developed a cloud-based ETL and data-integration platform to automate backups and data integrations from Podio. Utilized Node.js worker threads, BullMQ, and Redis for concurrent task execution at scale. Designed microservices for Kubernetes deployment and built real-time monitoring, alerting, operational dashboards, process logging, and usage tracking.
- **Tech Stack:** Node.js, TypeScript, BullMQ, Redis, Docker, Kubernetes, MongoDB, MySQL, GitLab, AWS

API Integrations

- Implemented secure and scalable B2B API integrations with QuickBooks, SAP Concur, Azure AD, and HubSpot. Built a PHP SDK for SAP Concur using PSR-7 to streamline integration across multiple clients. Configured a HubSpot-to-BigQuery data pipeline on GCP for analytics and collaborated with .NET and backend teams to develop REST APIs for legacy systems including MS NAV.
- **Tech Stack:** PHP, REST APIs, HubSpot API, QuickBooks, SAP Concur, Azure AD, BigQuery, GCP, OAuth 2.0

CVE Alert System

- Built a real-time vulnerability alert platform during a 24-hour solo hackathon. Integrated CVE feeds to monitor security issues affecting registered domains and technologies, implemented automated notifications and reporting for high-risk vulnerabilities, and developed a chatbot for custom security alerts.
- **Tech Stack:** TypeScript, Express.js, MongoDB, Azure BotBuilder, HTML/CSS/JavaScript, jQuery

Education

BSc Computer Science

Kerala University

05/2014 - 03/2017

Skills

- **Languages:** JavaScript, TypeScript, Python, PHP
- **Backend & Full Stack:** Node.js, Express.js, NestJS, Django, Flask, Laravel, React.js, Vue.js, jQuery
- **API & Integration:** RESTful APIs, GraphQL, API Gateways, API Design, API Integration, B2B Integrations, OAuth 2.0, Azure AD, WebSockets, gRPC
- **Architecture & System Design:** Distributed Systems, Microservices Architecture, Event-Driven Architecture, Message-Driven Architecture, Domain-Driven Design (DDD), Scalable Systems, High-Availability Systems, Fault-Tolerant Systems
- **Messaging & Streaming:** Kafka, RabbitMQ, Redis, BullMQ, Azure Event Hub
- **Databases & Data:** MongoDB, PostgreSQL, MySQL, MariaDB, BigQuery
- **DevOps & Infrastructure:** Kubernetes, Docker, Terraform, Jenkins, GitLab CI/CD, Infrastructure as Code (IaC), CI/CD, Automated Deployments
- **Cloud:** AWS, GCP, Azure, Cloud-Native Development
- **Monitoring & Observability:** Prometheus, Grafana, ELK Stack, Logging, Monitoring, Alerting, Performance Monitoring
- **Security:** OAuth 2.0, JWT, Azure AD, Secure API Development, Vulnerability Monitoring, CVE Integration
- **Engineering Practices:** Agile/Scrum, Code Reviews, Clean Code, Technical Reviews, Performance Optimization, Reliability Engineering, Technical Leadership, Team Mentoring

Additional Strengths

- Technical leadership
- Team leadership and mentoring
- Backend engineering
- Full-stack development
- Distributed systems
- API architecture
- API development
- API integration
- B2B integrations
- Microservices
- Event-driven architecture
- Message-driven architecture
- System design
- Performance optimization
- Cloud-native development
- DevOps and infrastructure automation
- CI/CD
- Kubernetes and containerization
- Observability and monitoring
- Reliability and fault tolerance
- Cross-functional stakeholder collaboration
- Agile/Scrum
- Agentic AI research and development
- Security-focused development
- Vulnerability monitoring
- ETL and data integration